

Does an Ability-Conditioned LLM Student Simulator Hold Up as a Construct?

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Abstract: This study examined whether a large language model can serve as a faithful, ability-conditioned student simulator that exhibits the full nomological network of a student construct. The simulator produces a student's answer, optionally preceded by its reasoning, to multiple-choice fraction comparison and equivalence problems, conditioned on an ability tier. We recast validity into four consistency evidences — profile–behavior consistency, fidelity of error, boundary of competence, and stateful learning — and tested them concurrently rather than one link at a time, with five conditions: zero-shot prompting and a 2×2 fine-tuning design (reasoning × history), against the real per-tier gold from the PSLC DataShop Fractions 2012 corpus. Fine-tuning with history and reasoning produced the best simulation, but the competence paradox persisted, most sharply at the middle tier; the reproduced misconception was consistently enacted within the trained problem distribution but did not generalize to unseen problems; competence was bounded by condition; and the learning mechanism was not reproduced, so stateful learning was not fulfilled. The contribution is a fully real-data nomological network and a four-condition causal ablation, stronger in data realism than prior prompted or part-synthetic simulators.

Keywords: Student simulation; Large language models; Construct validity; Fine-tuning; Learning analytics

1. Introduction

Educational innovation is costly in time and human-subject studies; simulated students offer a complementary methodology, supporting instructional testing (Yueh et al., 2026), data generation (Gao et al., 2025), teacher training and social learning (Jin et al., 2025). LLMs have boosted the research agenda in student simulation compared to the rule-based agents (Marquez-Carpintero et al., 2026). Both prompt-based (Lu & Wang, 2024; Yueh et al., 2026) and fine-tuned agents (Qi et al., 2026) have proved useful. Reasoning texts and trajectory history are effective for increasing validity of simulated students (Qi et al., 2026; Xu et al., 2025). Yuan et al. (2026) pointed out current limitation of simulated students: over-emphasize surface realism at the expense of epistemic fidelity — a tension rooted in the nature of LLMs, whose fluency can mask an unspecified or invalid learner state. We propose a set of behavioral evidences through a four-step learning cycle — theory, act, feedback, update — based on Neisser's (1976) perceptual cycle. Instrumenting the theory and act steps and the update between cycles, we link (1) profile-behavior consistency, (2) fidelity of error, (3) boundary of competence and; (4) stateful learning into a nomological network (Cronbach & Meehl, 1955). Each evidence closes one way in which fluent text can fake a learner that a surface-realism check misses: a profile worn as a style rather than a competence level (1), a plausible wrong answer without the misconception behind it (2), profiled incompetence leaking back to expert competence on unseen material (3), and a learning curve that echoes the input rather than a state change (4). Concretely, the simulator answers multiple-choice fraction

problems, optionally preceded by its reasoning, given an ability profile and, in some conditions, the student's prior trajectory.

While previous research (Lu & Wang 2024; Qi et al., 2026; Yueh et al. 2026) has examined the nomological network only partially, this question remains unanswered: Does an ability-conditioned LLM simulator exhibit a valid nomological network of student behavior, and which fine-tuning condition best realizes that nomological network? We examined the nomological behavioral evidences with fine-tuned LLMs under a 2×2 design (reasoning × history) against a zero-shot prompting baseline, on a single well-defined domain (Koedinger et al., 2010). Methodologically this study serves as a case of secondary use of shared, anonymized educational data — built and validated from years-old, de-identified DataShop logs — pointing to the validated simulator as itself a privacy-preserving, shareable synthetic-data artifact.

2. Literature Review

2.1 Forming a nomological network for simulated students

Large language models are the strongest behavioral substrate currently available for simulating human responses (Argyle et al., 2023), but student simulation faces a competence paradox (Yuan et al., 2026) a broadly capable model is asked to behave as a partial learner. Fluency makes surface realism cheap, so the binding question is not capability but validity — whether the simulated competence is the real thing or a convincing statistic.

We organize these evidences with a four-step learning cycle: a learner holds a working *theory of the task, acts on it, receives feedback from the environment, and updates the theory before the next attempt*. The formulation follows Neisser's (1976) perceptual cycle, in which a schema directs exploration, exploration samples the world, and the world modifies the schema. We promote that final modification to an element of its own, which lets the update be measured rather than assumed and avoids committing to schema theory. Instrumenting theory, act and update yields four pillars.

These serve as the coupled links of one construct that must converge jointly on real behavior (Cronbach & Meehl, 1955); fidelity of error and boundary of competence are adopted from prior validity criteria (Yuan et al. 2026). (1) Profile–behavior consistency: varying the ability must produce a correspondingly graded response, and within a response accuracy and error pattern must cohere — a struggling learner is not merely less accurate but wrong in a structured way. (2) Fidelity of error: a single misconception must drive the same mistake across isomorphic items, rather than memorized item-specific answers. (3) Boundary of competence: the response must stay within the profiled ability, neither leaking expert competence nor collapsing, and hold on an unseen problem set. (4) Stateful learning: ability evolves over practice, so the trajectory should show latent-state change — accuracy rising and the misconception decaying together — trackable as the learn-rate of Bayesian knowledge tracing and the per-opportunity slope of an additive factor model (Corbett & Anderson, 1995).

2.2 Prior simulators under the epistemic framework

Generative student simulators are built either by prompting with student profiles (Lu & Wang, 2024; Yueh et al., 2026) or by fine-tuning on learning logs (Qi et al., 2026). Qi et al. (2026) used LLM-generated reasoning on misconception error for fine-tuning, and gained a valid simulation against baseline, which complements the asymmetry between the correct data models are trained on, and the need to simulate student failure (Worden et al., 2026).

A separate line uses LLMs for student modeling / knowledge tracing, which predict a real student's answers from history rather than generating a student. Xu et al. (2025) fine-tune a correctness classifier on real course materials and interaction history, showing history is an effective conditioning signal for student simulation, and Worden et al. (2026) find that frontier

LLMs cannot trace an evolving knowledge state from raw history — an “optimism bias” over-predicts correctness, and accuracy does not improve with longer history.

Read against the four evidences, prior simulators cluster on the static links and thin out on the dynamic ones. Most validate the profile-behavior link against real students. Lu & Wang (2024) reproduce both facets: their generative students’ per-item scores correlate with 100 real students ($r \approx 0.72$), and a specified misconception recurs across items, driving the same wrong choice $\approx 82\%$ of the time on new questions (fidelity of error). Qi et al. (2026) obtain a monotone low, medium, high gradient with tier-differentiated error types. Yueh et al. (2026), by contrast, correlate ability only with their own simulated scores ($r \approx 0.34\text{--}0.54$), with no real-student criterion. What none establishes is a learned, over-time state: Qi et al.’s (2026) proficiency is a static per-session label, Lu & Wang (2024) deliberately suppress cross-item carry-over, and Yueh et al. (2026) adds only a hand-built error transition, so misconception structure and learning trajectory are never held together. Whether behavior stays within the profile on an unseen problem set is likewise rarely tested.

Table 1 scores each system on the four evidences: none covers the coupled links end-to-end, and none examines the two dynamic links, boundary of competence and stateful learning. This study targets exactly that gap, based on authentic student log data (Koedinger et al., 2010). We conducted a 2×2 (history, reasoning) ablation versus a prompted baseline, which isolates the conditioning lever that moves competence, and scored the competence boundary and learning trajectory against real students on two channels, accuracy rise and misconception decay.

Table 1. Coverage of the four behavioral evidences by prior LLM student simulators

Evidence	Description	1	2	3	4
Profile-Behavior	profile predicts accuracy	E	E	P	E
Fidelity of Error	one misconception recurs across similar items	E	P	N	E
Boundary of Competence	behavior stays within profiled ability	P	P	N	E
Stateful Learning	latent state evolves over practice	N	N	P	E

Note: a. E = designed and validated as holding against real human behavior; P = partially examined; N = not addressed. b. 1: Lu & Wang 2024; 2: Qi et al., 2026; 3: Yueh et al. 2026; 4: Current Study

3. Research Methods

3.1 Data and tasks

Data come from the PSLC DataShop Fractions 2012 lab experiment (Koedinger et al., 2010), authored in CTAT/Tutorshop (Aleven et al., 2006). The environment is a step-based tutor with immediate correctness feedback and hints, so a within-session learning trajectory is expected in the log, the premise Evidence 4 relies on. We recover the per-item correspondence from the tutor’s CTAT tracing graphs, with correctness read from DataShop’s Outcome column, yielding 57,734 graded responses from 176 students (342 per-student trajectories), fully real and per-item grounded. Eight problems carry an evaluable decisive multiple-choice step and define two tasks. Comparison (step combo1_3): “Is a/b larger, smaller, or equal to c/d ?” — the correct rule is smaller denominator $\Rightarrow$ larger, and the wrong-direction distractor is the whole-number-bias misconception. Equivalence (step fract4_denom0, the most discriminating): rewrite c/d by multiplying; distractors are the real top wrong values (additive error). Ability tiers are terciles of each student’s overall fraction accuracy (59 / 58 / 59 low / medium / high). Throughout, the real per-tier gold — the criterion every condition is scored against — is the held-out real students’ accuracy of that tier at the decisive step.

3.2 Conditions and fine-tuning

All five conditions share Qwen2.5-7B-Instruct and the same ability-conditioned system prompt, and every trained target ends in ANSWER. The four fine-tuned conditions cross two factors (Table 2): reasoning (narr), whether an authored misconception-enacting reasoning text precedes the answer, and history (hist-), whether the student’s real prior trajectory is carried in the training example’s user turn.

Table 2. Conditions; the short label in parentheses is used in the figures and in §4.

Condition	Manipulation	Post Training
base	ability prompt only	none
answer-only (ans)	frequency-weighted to the real per-tier distribution.	fine-tuned; target = bare ANSWER
reasoning (narr)	Claude Opus 4.8 augments tier-appropriate reasoning that enacts the misconception	fine-tuned; target = misconception reasoning then ANSWER.
history (histans)	prompt carries the student’s prior-step record (problem step=response)	fine-tuned; target = bare ANSWER.
history + reasoning (hist)	prompt carries the real trajectory	fine-tuned; target = misconception reasoning, then ANSWER

The reasoning texts, authored by Claude Opus 4.8, are the only generated content in the pipeline: each verbalizes how a student reaches an answer they actually gave. Keyed by (task, problem, answer), the corpus is the complete cross-product of the 8 problems and the answers real students produced — 33 texts: the 4 comparison problems each admit 3 closed options (12), and real students gave 5–6 distinct values on each of the 4 open equivalence problems (21) — reused across the training corpus and audited as a census. The training pool is always the students not in the held-out unit, with the held-out problem/task additionally hidden from the history prefix. The two families differ in size by an order of magnitude because they are built on different units: the no-history conditions aggregate real responses per (problem, tier), frequency-weighted to the real per-tier distribution (244 records); the history conditions use one example per real student-response (1,689), its user turn carrying that student’s preceding ≤ 12 logged steps.

3.3 Measurement

Each evidence is scored against real logged behavior with its own held-out unit: Evidence 1 uses 5-fold student cross-validation (CV); Evidence 2 leave-one-problem-out (LOPO: unseen item, seen task); Evidence 3 leave-one-task-out (LOTO: unseen task, each direction reported separately); Evidence 4 replays the fold-0 students longitudinally; regimes are never pooled. Sampling is at $T = 1.0$: the no-history conditions generate 480 responses per fold (8 problems $\times$ 3 tiers $\times$ 20 simulated students), the history conditions 6 (fold 0) or 4 (folds 1–4) per held-out student-response carrying that student’s real trajectory (1,488–2,008 per fold). Per cell we score accuracy (correct / parsed); from the held-out students’ own answer counts we define the misconception as the modal real wrong answer, all other wrong answers pooled as other. Distributional fits use the bias-corrected and accelerated (BCa) bootstrap: per-student accuracy by W_1 , the earth-mover distance to real, and cross-problem consistency by the model/real SD ratio; with $n = 9/13/8$ students per tier, non-significance means indistinguishable at this power, not identical.

3.3.1 Profile–behavior consistency

Resulted from 5-fold student cross-validation, this evidence examined whether per-tier accuracy is correctly graded low<medium<high against the real gold, and whether the error coheres with the profile. Coherence is examined at the reasoning layer with a semantic classifier that labels each wrong generation as an enacted misconception (the reasoning enacts the wrong answer) or a referent slip (correct reasoning, mis-reported answer): for comparison, by whether the model’s final comparative claim is mathematically false

(misconception) or true (slip); for equivalence, by whether the method it states is additive / keep-denominator (misconception) or correct multiplication (slip).

3.3.2 Fidelity of error

This evidence examined whether one misconception drives the same wrong answer on an unseen problem of the same task — a transferred competence representation rather than memorized per-problem answers.

3.3.3 Boundary of competence

The evidence was derived from leave-one-task-out (train comparison, test equivalence, and vice versa — an unseen task) together with a base-model prompting control: whether behavior stays within the profiled ability or the competence floor can be crossed by conditioning.

3.3.4 Stateful learning

For this evidence, we ordered the held-out (fold-0) students' decisive-step responses by practice opportunity, fed the history conditions the real trajectory step by step, and scored the within-student last-third – first-third change on two channels — accuracy (should rise) and misconception rate (should decay) — for students with at least three opportunities on the task (27 comparison, 29 equivalence). Simulated and real sequences share students, so both sides are resampled together in a student-clustered bootstrap (5,000 draws) and every comparison is reported as the simulator-minus-real difference with a 95% interval. The real target comes from criterion knowledge-tracing models on the real responses (2,238 first attempts, 173 students; one knowledge component (KC) per task): a fixed-effects additive factor model (AFM) supplies the two-channel per-opportunity slopes, corroborated by a standard Bayesian knowledge tracing (BKT) learn-rate, with a misconception-aware BKT confirming the misconception is state-organized.

4. Results

4.1 Profile-Behavioral Consistency

As shown in Figure 1a, the base ignores the ability prompt, remaining over-competent on both tasks (equivalence-low 1.000 vs real 0.405): it controls style, not competence. Answer-only barely moves, its correctness prior dominating (comparison-low 0.805). History is the dominant lever — the real trajectory pulls the low tier toward the floor (comparison 0.688 [0.63, 0.75], equivalence 0.795 [0.71, 0.88]) while no-history conditions stay at 0.81–0.88. Reasoning is a task-dependent interaction — negligible on comparison, but on equivalence it helps and only with history (to 0.662 [0.56, 0.77], the largest drop), making history + reasoning the best condition. Fold-to-fold variability is small (95% CI half-width $\leq \approx 0.10$, below the history effect ≈ 0.14 – 0.19) and every interval clears the real floor. The ordering holds on the distribution too (Figure 1b), but unevenly: on equivalence the history conditions are statistically indistinguishable from real at every tier (history + reasoning W_1 0.07 / 0.11 / 0.05; permutation $p = 0.59$ / 0.16 / 0.61), whereas on comparison even the best condition stays detectably over-competent at the low and medium tiers ($p \leq 0.02$), and the no-history conditions differ almost everywhere (W_1 0.25–0.51). The medium tier is hardest to match: every condition collapses the real students' broad 0.50–1.00 mixture into a narrow, over-competent lump.

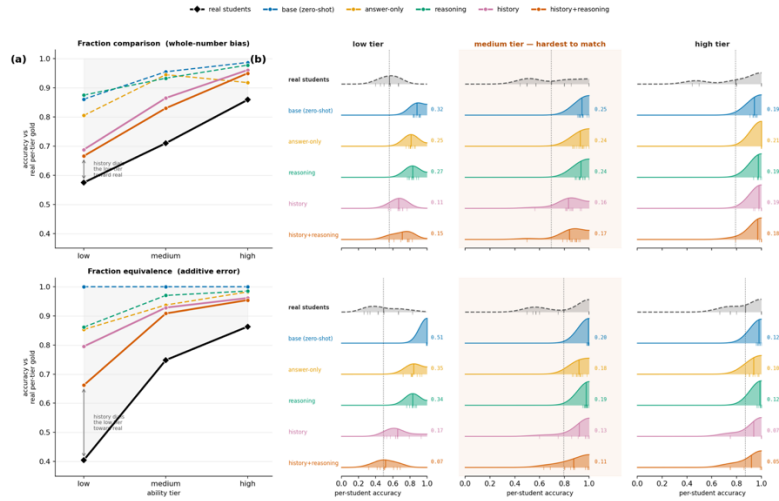


Figure 1. Profile-behavior consistency under 5-fold student cross-validation. (a) Per-tier accuracy of the five conditions against the real per-tier gold (black diamonds; solid = history conditions, dashed = no-history); a faithful condition tracks the black line rather than sitting above it. (b) The per-student accuracy distributions behind (a), by task (rows) $\times$ tier (columns), one ridge per condition against the real distribution (grey), annotated with its W_1 distance to real (smaller = closer).

A struggling learner should be wrong in a structured way — its errors the profiled misconception, not noise. At the answer layer they are: induced wrong answers land on the misconception option (rate ≈ 1.0), and the real tier gradient of the misconception share is reproduced. It holds at the reasoning layer too (Figure 2a): labelling each wrong generation by the truth of the model's own final claim — an enacted misconception if false, a referent slip if true but predicated of the other fraction — enacted misconceptions run 0.73–0.76 on comparison (narr) and 0.99–1.00 (hist), and 0.84–0.92 on equivalence (narr) and 0.42–0.63 (hist), among classified errors. Counting every unscorable generation (0–25% per cell) against the misconception, it remains the majority route in 11 of 12 cells. Because the training reasoning texts were authored by Claude Opus 4.8, this coherence shows that fine-tuning transferred a consistently enacted misconception pattern, not an independently held belief.

4.2 Fidelity of Error

The misconception's answer recurs — and on the problems the model has seen it recurs consistently: pooling the four isomorphic items across tiers, its cross-problem spread is no larger than real students' (SD ratio 1.30 $\times$, BCa 95% CI [0.79, 2.47] on comparison; 0.65 $\times$ [0.45, 0.98] on equivalence). Consistency breaks only in generalization to unseen problems (Figure 2b): on leave-one-problem-out the simulator becomes significantly more variable across problems than real — SD ratio 3.07 $\times$ [1.84, 5.98] on comparison and 1.99 $\times$ [1.24, 3.37] on equivalence, both intervals excluding 1 — and drifts off the real level. The misconception is thus applied with real-student stability within the trained problem distribution, but does not carry that stability to novel isomorphic items.

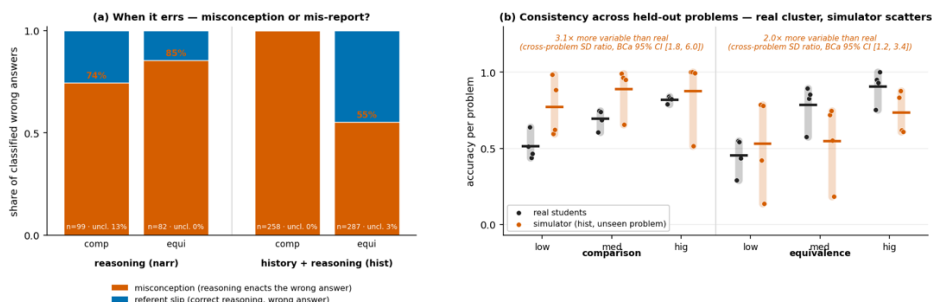


Figure 2. The simulator's errors at the reasoning and problem levels. (a) Error coherence for narr and hist: whether each wrong generation's reasoning enacts the misconception or states the correct method and mis-reports the answer; shares are among classified errors

only (n and unclassified share per bar), pooled over tiers. (b) Leave-one-problem-out: accuracy on each held-out problem (the same correct-over-parsed accuracy as Figure 1), real students versus hist, with the cross-problem SD ratio and its BCa 95% CI.

4.3 Boundary of Competence

Leave-one-task-out transfer is a two-instance demonstration, one held-out task per direction, and it is asymmetric. Training on comparison and testing on equivalence leaves every condition at ceiling (base, ans, histans 1.00; narr ≥ 0.99 ; only hist off it at 0.960–0.968, against real 0.405 / 0.748 / 0.863): whole-number-bias training cannot induce the additive error. The reverse direction shifts competence down but destroys the tier gradient (narr flat at 0.69 / 0.69 / 0.71; ans and histans invert the high tier to 0.64 / 0.63; real 0.575 / 0.710 / 0.859). Conditioning thus generalizes to an unseen problem within a task but not to an unseen task: the specific error process is task-bound and needs in-task supervision.

A base-model control confirms that prompting cannot cross the competence floor even when handed the real trajectory, so the history conditions' competence-lowering is a fine-tuning effect, not an inference-time artefact. No condition crosses the real low-ability floor (the best stays +0.09 above it on comparison and +0.26 on equivalence): faithful simulation is windowed, and the medium tier is the fuzziest (Figure 1b), a broad mixture of mastery states that a statically conditioned model collapses to a narrow, over-competent lump.

4.4 Stateful Learning

The criterion is two-channel: a faithful trajectory must both gain accuracy and shed the misconception, as real students do (Figure 3). On the comparison task — in-window, i.e. inside the base model's competence range in the sense of §4.3 — the point estimates are consistent with that bar, though with 27 held-out students the result is suggestive rather than confirmatory. Fed the real history, hist is the only condition whose two channels are both indistinguishable from real students': accuracy +0.199 against +0.191 (difference +0.008 [−0.098, +0.117]) and misconception −0.030 against −0.062 (+0.042 [−0.042, +0.121]); the four other conditions differ on both channels with intervals excluding zero.

The comparison result is nonetheless weak evidence, because the real decay it must match is itself small, its interval covering zero (−0.062 [−0.141, +0.009]). Equivalence is the discriminating task: there the real decay is established (−0.103 [−0.175, −0.035]) and no condition reproduces both channels — ans and histans raise accuracy only as a frequency echo of the injected answer distribution (+0.142 / +0.158 vs real +0.319), and the same hist that learns on comparison drives its equivalence misconception the wrong way (+0.089, difference +0.195 [+0.108, +0.279]). Matching accuracy while the misconception moves wrong is the signature of a re-described history, not an enacted state transition.

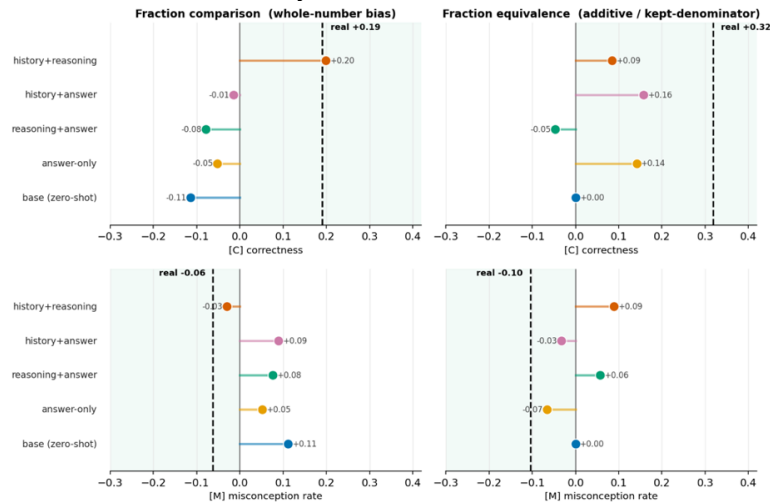


Figure 3. Stateful learning: within-student last-third – first-third change in correctness [C] and misconception rate [M] for each condition (points), against the real students' change (dashed line); a faithful condition must match both.

A fixed-effects AFM corroborates the target: within students, correctness and the misconception carry opposite-signed per-opportunity slopes (comparison $+0.13$ / -0.12 ; equivalence $+0.20$ / -0.06), and a misconception-aware BKT fits the three-way outcome far better than a stateless baseline — the misconception is a real, state-linked channel.

5. Discussion

5.1 Key Findings

In brief, an ability-conditioned LLM simulator does exhibit the nomological network of student behavior, but only partially (Table 3). We aggregate the results into three findings: (1) fine-tuning with history and reasoning produces the best simulation; (2) the competence paradox persists, most sharply at the middle tier; (3) the misconception is consistently enacted in the model’s reasoning on trained items, but its hold does not generalize like a student’s.

Table 3. Evidence $\times$ condition (E achieved / P partially achieved / N not achieved); hist reaches the two-channel bar only on comparison and only suggestively.

Evidence	base	ans	narr	histans	hist
Profile–behavior consistency	N	P	P	P	E
Fidelity of error	N	N	P	N	P
Boundary of competence	N	N	N	P	P
Stateful learning	N	N	N	N	P

Of the four levers, only the real trajectory moves competence, making history + reasoning the single best condition. Prompting controls style, answer-only inherits the correctness prior, and the misconception reasoning helps only on equivalence and only with history. History + reasoning dials the low tier toward the real floor (comparison 0.666, equivalence 0.662), enacts the misconception in its reasoning on both tasks, and reproduces the two-channel trajectory on comparison. Examined jointly, these are the coupled links of one nomological network (Cronbach & Meehl, 1955) scored against real per-item data — the first end-to-end examination of the whole network, which no prior simulator attempts.

Because these tasks sit inside the base model’s competence window, conditioning moves competence toward but not to the real low-ability floor (§4.3): faithful simulation is windowed. This refines Qi et al. (2026): a monotone gradient is a claim about ordering, not calibration — a simulator can match low<medium<high while every tier floats over-competent — which a partly synthetic low-tier target cannot expose; it mirrors Worden et al.’s (2026) knowledge-tracing “optimism bias”, the prediction-side counterpart of our generation-side over-competence. The paradox is sharpest at the middle tier (best $+0.12$ / $+0.16$ while low and high align).

Scored by the truth of its own claim, the simulator’s errors are enacted misconceptions in 11 of 12 cells: it mostly holds the wrong ordering and says so, rather than reasoning correctly and mis-reporting. It applies that pattern consistently across the isomorphic items it was trained on — its per-tier accuracy across those problems is as stable as real students’ (SD 0.04–0.09 vs 0.02–0.13) — but not to problems it has not seen: under leave-one-problem-out its cross-problem spread is two to three times real students’ (SD ratio $3.07\times$ [BCa 95% CI 1.84, 5.98] on comparison, $1.99\times$ [1.24, 3.37] on equivalence) and it drifts over-competent. The misconception is thus enacted item by item within the trained problem distribution, not carried as the skill-general disposition a struggling student brings to any new instance of the same skill.

The simulator matches the output distribution of learning but not its process. In-context conditioning updates no weights: the model reads the successes and errors off the injected trajectory to set a competence level for its next answer, rather than resolving a misconception and carrying that resolution forward. The two-channel score exposes this — the condition that matches the accuracy rise on comparison drives its misconception the wrong way on equivalence — and the same limit is why competence set by conditioning neither persists to a held-out problem nor evolves as a real latent state would: it is set anew by each context,

never consolidated into the weights. A surface-valid learning curve is therefore weak evidence for genuine learning: we reproduce learning's behavior, not its mechanism.

5.2 Limitations

Across the four-step cycle, this study instruments theory, act and update, but the feedback step is replayed from logs rather than produced interactively: the simulated learner reacts to a recorded tutor. Its results therefore speak to uses that consume a learner's responses — data generation and item evaluation — and not to uses that require interaction, such as teacher training or peer learning. On the other hand, a single well-defined domain (Koedinger et al., 2010) is only a fraction of various problem solving, compared to cross domain problems, and even ill-defined problems. Finally, we use Qwen2.5-7B and do not replicate Simulacra's (Xu et al., 2025) frontier-scale (GPT-4o / GPT-4o-mini) results.

Two limits are internal to the data construction. The equivalence data is imbalanced: because the correct answer is always the most frequent, the discarded tail is almost entirely wrong answers, leaving the low-tier equivalence target 0.117 too easy on the accuracy scale (a reweighting estimate: 9–16% of the equivalence over-competence), while comparison, a closed three-option step, is unaffected. Second, the reasoning-layer classifier leaves 0–25% of wrong generations unclassified, so misconception shares are reported among classified errors with their n . Three assumptions also deserve flagging: the stateful-learning verdict rests on 27–29 students from one fold and is suggestive at best; the modal wrong answer is equated with the target misconception, which distractor structure licenses but cannot prove; and tier-level conditioning assumes within-tier misconception homogeneity, an assumption the medium-tier mixture already strains.

Each limitation is also a direction: the feedback step can be made interactive, the update step instrumented directly, the domain widened to ill-defined and cross-domain tasks where the misconception inventory is less crisp, and the base scaled to test whether the competence window shifts rather than closes — scale should work against low-ability fidelity, since a stronger base is more over-competent, so the floor is the quantity to track.

6. Implications and Conclusion

The findings translate into practical guidance for building an ability-conditioned simulator. First, match task and base: the floor is a joint property of task and base, so choose the base for the intended competence rather than assume it; a stronger base is more over-competent. Second, fine-tune on real trajectories with an explicit misconception reasoning text rather than prompting: the text installs a misconception the model enacts in its reasoning, so author one for each misconception to be enacted, but expect it to be applied item by item rather than carried across problems. Third, validate beyond the answer distribution, particularly across problems, where a surface per-tier match hides that the misconception does not hold as a disposition. Fourth, do not read learning dynamics off an in-context simulator without an architectural memory. These are conditions for a usable instrument, not a route to a valid learning mechanism: in-context simulation cannot be patched into stateful learning. Within these bounds, the simulator is a practical data-generation instrument: cold-starting a tutor's knowledge-tracing priors and policy tuning before real students arrive; stress-testing whether an item bank's distractors capture the intended misconception and order items by difficulty; oversampling the low-tier cells that real logs leave sparse; and generating reasoning-annotated error trajectories for training diagnostic classifiers and for teacher preparation — each inside the window, at the answer-distribution layer, never as learning dynamics. Being validated against shared, de-identified logs, the simulator is itself a shareable, privacy-preserving artefact that carries the real misconception structure without exposing any learner's record.

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